

Computer
Programming
ECEg1052
Cosc 2111_

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Computer Programming

- Process of writing, testing, debugging, and maintaining source code of computer programs.
- Problem: gap between existing and desired/proposed situation.
- Problem solving techniques:
 - Analyze problem
 - Outline problem requirements: UI? Data? Output?
 - Design algorithm

Computer Programming

- Properties of algorithms:
 - Correct: always desired output
 - Unambiguous: clearly defined steps
 - Precise: simple and efficient
 - Efficient: measured in time and space
 - Finiteness: terminates

Computer Programming

- Tools to document programs:
 - Flow charts: represent processes with symbols
 - rectangle: process
 - skewed rectangle: input/output
 - diamond: decision (if/else)
 - circle: connector
 - rectangle with side stripes: subroutine call
 - rounded rectangle: terminal (start/stop)
 - lines
 - Structured chart: logical functions
 - pseudocode

Computer Programming

- Programming language: defined by syntax and semantics
 - Low level languages: machine-dependent
 - High level languages: machine-independent
- Programming paradigms: programmer's view of code execution
 - Procedural: list of operations to reach desired state
 - Structured (in Procedural): break program structure to pieces (no GOTO statements)
 - OOP: specify data structure and operations to be applied on those data structures

Computer Programming

- C++ elements
 - Keywords: unique meaning in C++, cant be reused for other purpose
 - Identifiers: name of function or data
 - Literals: constant values i.e. numbers or strings (10.5)
 - Comments: descriptive text
 - Data types, Variables, Constants
 - Console communication (cin, cout)

Computer Programming

- C++ elements
 - Operators: Assignment:
 - Assignment Operators (=)
 - Arithmetic Operators (+, -, *, /, %)
 - Relational operators (==, !=, <, <=, >, >=)
 - Logical operators (! Negation, &&, ||)
 - Increment/decrement Operators
 - Ternary /Conditional operator (? :)

Computer Programming

- C++ elements
 - Control statements: flow control to determine what gets executed
 - Declaration statements: define variables
 - Assignment statements: algebraic computations
 - Branching statements: alternate paths
 - Loop statements: repeated computations
 - Scope of variables
 - Local variable: inside a block, has precedence if same name with global variable
 - Global variable: available from any part

Computer Programming

- C++ elements
 - Pointer: holds memory address of another variable
 - Reference operator &: address of
 - Dereference operator *: value pointed by

Computer Programming

- C++ elements
 - File Management (fstream library)
 - File: used to store data in storage devices permanently
 - **ofstream**: Stream class to write on file
 - **ifstream**: Stream class to read from files
 - **fstream**: Stream class to both read and write from/to files.

Computer Programming

- C++ elements
 - File Management (fstream library)
 - File modes:
 - `ios::in` - opens and reads if file is found
 - `ios::out` - opens and writes, if found overwrite else new file is created
 - `ios::app` - opens and appends if found, else new
 - `ios::binary` - open if found for binary inp/output
 - `ios::ate` - open and position at end of file, used with `ios::in`, `ios::out` and `ios::binary` for appending
 - `ios::trunc` - open and truncate/delete all content if found
 - `ios::nocreate`: dont create new file if not found
 - Combine with pipe (|) after file name
 - example: `fstream writefemp;`
`writefemp.open("Employee.txt",ios::app|ios::out)`

